A Missing Details from Section 7: Related Work

Our problem connects ranked query processing, learning to rank, ranking explanation, rank distances, and linear inequalities. We seek one common score on the supplied features that approximates a complete target order. The following discussion explains how this objective differs from those studied in each of these areas.

A.1 Ranked retrieval and utility queries

Fagin et al. developed algorithms for combining scores through sorted and random access [27], and surveyed the broader problem of combining fuzzy information [25]. These methods efficiently evaluate an aggregation rule; they do not choose a rule to reproduce an externally supplied order. The onion technique uses geometric layers to index linear optimization queries [12]. RankingCube organizes multidimensional data to answer ranking queries [65], while Das et al. answer top-k queries by combining information from materialized views [17]. Reverse top-k queries ask which weight vectors place a designated tuple among the top results [62]. All these formulations connect weights to ranked query answers. Our objective instead evaluates the entire induced order against a target by counting reversed pairs or summing changes in item positions.

Other work addresses uncertainty about user preferences. A *skyline* contains tuples for which no other tuple is at least as good in every relevant attribute and strictly better in at least one [7]. *Regret-minimizing representative sets* select a small subset whose best item remains close to the best item in the full data across a family of utility functions [45]. *Rank-regret representatives* measure this loss by rank rather than utility [6]; Shetiya et al. give a unified optimization approach for several representative-set problems [56]. These methods choose items, whereas we keep every item and choose the scoring rule.

Interactive approaches obtain additional preference information from the user. Nanongkai et al. study interactive regret minimization [46]; Qian et al. learn preferences through adaptive pairwise comparisons [50]; Wang et al. search interactively for one of a user's top-k items [63]. Our input already specifies a full order. Our objective measures how closely a fixed linear feature representation can realize that order; it neither selects follow-up questions nor returns a subset of items.

A.2 Learning and explaining linear rankings

Learning-to-rank methods estimate scores from relevance judgments, clicks, or pairwise preferences. Joachims uses clickthrough data to learn pairwise ranking preferences with a support-vector method [32]. RankNet learns from pairwise comparisons using a differentiable probabilistic loss [8]. Sculley studies scalable learning-to-rank methods [54], and specialized algorithms improve the efficiency of linear ranking support-vector machines [2, 37]. These methods optimize training losses rather than the exact number of reversed pairs for one complete target order. For example, RankSVM penalizes both comparison errors and the size of the weight vector, while RankNet uses a smooth probabilistic loss.

Classical ordinal regression fits ordered responses through linear programming [59]. Metzler et al. directly optimize rank-based retrieval metrics using a numerical search over scoring parameters [42]. Shen et al. study linearized maximum-rank-correlation estimation [55]. These works motivate optimizing ranking quality directly, but their statistical or numerical guarantees differ from our worst-case approximation guarantees for a fixed finite input. We do not claim that optimizing a smooth training loss solves the minimum-disagreement problem.

A second line of work explains rankings or synthesizes scores. Gale and Marian study feature-contribution metrics for *monotonic ranking functions*, whose scores do not decrease when an attribute improves [29]. RANKHOW is especially close to our setting: Chen et al. fit a supplied ranking by minimizing positional error, allow weight constraints and score ties, and give an optimization model that combines real-valued weights with discrete ordering decisions. Their symbolic-gradient method searches for weight changes that improve the induced order [13]. Their exact region-enumeration bound has an exponent that depends on the number of features, so its polynomial-time conclusion assumes a fixed feature count. They also discuss inversion-based errors, including Kendall's Tau distance; their principal optimization formulation uses positional error. We instead give approximation guarantees when the feature count grows and require the linear score to induce the full ranking without ties. Why-not analyses identify changes that would place a missing item in a top-k answer or explain reverse top-k query results [30, 38]. RANKHOW directly optimizes ranking error; the why-not formulations focus on a designated item or subset. Our results complement these systems with worst-case guarantees and dimension-dependent lower bounds.

Feature-attribution methods quantify how an existing model's inputs contribute to its predictions. Ranking-specific methods include LIRME's local explanations [61], Rank-LIME [15], RankSHAP [16], and RankingSHAP [31]. We instead receive only feature vectors and a target order, without access to the model that created it. Yan et al. study exact explanations that augment a linear score with hidden group bonuses [66]. We keep the feature vectors unchanged and allow a positive optimum distance; our score has no bonus variables. Our algorithms and reductions use this fixed family of scores throughout.

A.3 Polynomial scores and nonlinear utilities

Polynomial scoring has a long history in information retrieval. Fuhr maps query–document pairs to feature vectors and fits polynomial estimates of relevance probabilities by minimizing squared error [28]. This estimates a numerical response from relevance feedback. Our polynomial extension instead receives a complete target order and optimizes its pairwise or positional disagreement; it needs neither relevance probabilities nor target score values.

Polynomial kernels provide another route to nonlinear ranking models. A *kernel* evaluates an inner product in a transformed feature representation without explicitly constructing its coordinates. More specifically, Joachims's SVM-rank documentation supports the polynomial kernel $K(x, z) = (s\langle x, z\rangle + c)^p$, with configurable degree p, scale s, and offset c [33]. Expanding this expression gives products of feature coordinates; with positive s, c, its feature representation includes the monomials up to degree p, with fixed rescaling. Raykar et al. explicitly compare linear RankSVM with a quadratic variant using $K(x, z) = (\langle x, z\rangle + c)^2$ in their experiments [52]. Their main algorithm uses a differentiable lower bound

on a generalized Wilcoxon–Mann–Whitney statistic, which measures how often items from ordered groups appear in the correct relative order. The quadratic RankSVM is a benchmark, not a theorem giving an approximation ratio for fitting a ranking with an unrestricted quadratic score. Ranking support-vector methods trade a penalty on comparison errors against a penalty on the score parameters. These training objectives differ from minimizing the exact ranking distance over all polynomial coefficients on a fixed input.

The database literature also considers nonlinear utility functions. Kessler Faulkner et al. study utilities that model diminishing returns, risk preferences, and substitution between attributes [35]. Qi et al. study multiplicative utilities, including Cobb–Douglas and constant-elasticity-of-substitution families [49]. Cobb–Douglas utilities multiply weighted powers of positive attributes; constant-elasticity-of-substitution utilities use a power aggregate to model substitution between attributes. These papers choose a small representative subset whose best item loses little utility relative to the best item in the full database. We keep every item and choose a common score to approximate the entire supplied order. Their regret bounds therefore address a different output and objective from our ranking-distance guarantees.

Our use of monomial features is standard, and we do not claim a new polynomial scoring family. Chen et al. also explicitly discuss polynomial-derived attributes for RANKHOW [13]. Appendix G uses that representation to prove deterministic $(r_p + 1)$- and $2(r_p + 1)$-approximations for minimum Kendall's Tau and footrule disagreement, respectively, where r_p is the effective dimension of the expanded features. It also proves a randomized maximum-agreement scheme that is FPT in r_p and the accuracy parameters, and a PTAS when $r_p = O(\log n)$. These are multiplicative guarantees relative to the best unrestricted polynomial score of each fixed maximum degree, rather than guarantees for relevance estimation or a regularized training loss.

The hardness extension needs more than feature expansion. Every linear score is a polynomial score, but allowing higher-degree terms may improve the optimum on an instance that is hard for linear scores. Our separate block construction converts *any* ranking induced by a polynomial score into weights for the original inequalities. This conversion increases the approximation factor by at most a constant that depends on the fixed degree. Theorems G.7 and G.8 consequently retain strong minimum-disagreement inapproximability as a function of item count, original dimension, and expanded effective dimension. These are computational lower bounds for fitting a complete order, not lower bounds on the utility loss of a representative subset. They concern arbitrary dimension; we do not claim that this reduction extends Theorem E.2 to higher degrees in logarithmic dimension.

A.4 Rank distances and aggregation

Kendall's Tau distance counts reversed pairs, while Spearman footrule adds the positional movements of the items. Diaconis and Graham show that footrule distance lies between Kendall's Tau distance and twice that distance [19]. We use this theorem only to transfer the approximation upper bound. Our paired-point construction proves an exact factor-two identity for its own instances; that identity is stronger than what the general metric comparison alone would give.

Rank aggregation combines several input orders into a consensus. Dwork et al. study aggregation for Web rankings [23], and Davenport and Kalagnanam investigate algorithms for the Kemeny rule, which minimizes pairwise disagreement with a collection of orders [18]. We receive one target, not a collection of votes. Moreover, we cannot choose an arbitrary permutation: the feature vectors restrict which orders a common linear score can produce. The difficulty therefore comes from realizability, not disagreement among several targets. Complementing the pair count equates exact maximum agreement and minimum Kendall's Tau disagreement, but does not equate their multiplicative approximation guarantees.

A.5 Geometry, feasible comparisons, and hardness

The weights that tie two items form a flat boundary in weight space, called a *hyperplane*. Formally, this boundary satisfies $\langle w, x_i - x_j \rangle = 0$. The collection of these boundaries forms a *hyperplane arrangement*. Removing them divides the remaining space into connected regions, called *cells*; every pair keeps its sign within a cell, so the ranking stays unchanged. This geometric description is classical [24, 51, 60]. Work on linear programming in fixed or low dimension provides another reason to treat dimension as a structural parameter [11, 41]. Our algorithms do not enumerate all these regions. The minimum-disagreement algorithm uses small incompatible sets of comparisons. The agreement scheme samples comparisons and uses a uniform estimation theorem for linear comparisons [4]; Appendix D.1 explains the relevant terms.

Orienting comparisons by the target gives strict homogeneous linear inequalities. Deleting as few inequalities as possible to obtain feasibility is equivalent to minimizing Kendall's Tau disagreement. This connects our objective to maximum feasible subsystems and minimum unsatisfied linear relations [3, 14]. The ranking setting adds structure: comparisons arise from all pairs of one point set, and the output must be a full strict ranking. Our reduction explicitly preserves this structure and proves exact relations for both distances and agreement. The strongest approximation lower bounds use the homogeneous inequality construction of [44], with the stronger exponents conditional on the Projection Games Conjecture [43]. For logarithmic ambient dimension, Appendix E.3 gives a separate reduction from k-Clique; the standard parameterized-complexity framework explains the assumption used in this result [22].

B Missing Details from Section 2

We use the notation of Section 2: $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ is the set of distinct items, π is the target ranking, C is its set of $M = \binom{n}{2}$ target comparisons, and Δ_e is the comparison vector for e. Recall that $r = \dim \operatorname{span}\{\Delta_e : e \in C\}$ is the effective dimension of X.

B.1 Explicit tie-breaking with rational weights

The algorithms must output a strict ranking even when a feasibility test leaves some scores equal. We prove Lemma 2.2 by giving an explicit polynomial-time construction that keeps every required comparison positive.

Proof. Assume $n \geq 2$, so every Δ_e is nonzero and $d \geq 1$. For an integer t, let $v(t) = (1, t, t^2, \ldots, t^{d-1})$. For each pair e, the function $\langle v(t), \Delta_e \rangle$ is a nonzero polynomial in t of degree at most $d-1$. It has at most $d-1$ distinct real roots. Thus all M pairs together forbid at most $M(d-1)$ integer values. Test the $M(d-1)+1$ integers from 0 through $M(d-1)$; at least one gives a vector v with no tied pair. When $d = 1$, the single choice $v = (1)$ already works. These integers and their powers have polynomial encoding length, and we test every inner product by rational arithmetic.

Let w be the supplied rational solution of $\mathcal{L}(S)$, and set $B = \max_{e \in C} |\langle v, \Delta_e \rangle|$. Define $U = 1/[2(1+B)]$. For any $0 < \alpha \leq U$ and $e \in S$, we have $\langle w + \alpha v, \Delta_e \rangle \geq 1 - UB > 1/2$. Thus the perturbation keeps all required comparisons.

For any pair, the equation $\langle w + \alpha v, \Delta_e \rangle = 0$ has at most one solution in α, because its coefficient $\langle v, \Delta_e \rangle$ is nonzero. Test the $M+1$ rational choices $\alpha_j = jU/(M+1)$, $j \in [M+1]$. Only M of them can be forbidden, so one makes all scores distinct. The arithmetic operations and rational encoding lengths remain polynomial in the input and in the supplied w. This proves both the preservation and running-time claims. For $n \leq 1$, any weight vector suffices. □

The hardness proofs also need this property for weights that may be real rather than rational. Given finitely many nonzero rational coefficient vectors, we can perturb a real weight vector so that every positive inner product stays positive and no inner product equals zero. Choose a direction whose inner product with every coefficient vector is nonzero, as in the proof of Lemma 2.2. A sufficiently small positive step in this direction preserves the finitely many positive inner products. Each coefficient vector rules out at most one step size, so a suitable size exists. The perturbation may satisfy additional inequalities, but it cannot turn a satisfied inequality into a violation.

C Missing Details from Section 3

C.1 Comparison deletion and Kendall's Tau disagreement

We prove Lemma 3.2 by relating deleted comparisons to reversed pairs. In one direction, deleting every reversed pair leaves a feasible comparison system. In the other, any feasible deletion set gives a ranking that reverses no more pairs than were deleted.

Proof. For any tie-free weight vector w, let $D_w = \{e \in C : \langle w, \Delta_e \rangle < 0\}$ denote the target comparisons reversed by σ_w. The remaining comparisons all have positive score differences. By Lemma 2.1, their comparison system is feasible, so D_w is a feasible deletion set. It has exactly $d_{\mathrm{KT}}(\pi, \sigma_w)$ elements. Applying this observation to an optimal ranking gives $\mathrm{OPT}_{\mathrm{del}} \leq \mathrm{OPT}_{\mathrm{KT}}$.

Conversely, take any feasible deletion set D and solve $\mathcal{L}(C \setminus D)$. Its solution puts every undeleted pair in the target order with score difference at least 1. It may tie some deleted pairs, but Lemma 2.2 breaks those ties while keeping every undeleted comparison positive. The resulting ranking can reverse only pairs in D, so its Kendall's Tau distance is at most $|D|$. Apply this construction to a minimum deletion set to obtain the reverse inequality. Together, the two directions prove equality. □

C.2 Implication for Spearman footrule

Spearman footrule distance charges item movements rather than inverted pairs, so the Kendall's Tau deletion analysis does not apply to it directly. The Diaconis–Graham inequalities in (2) relate the two costs. We run the same algorithm, with no change to its input or pseudocode.

Corollary C.1. *Given $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{R}^d$ and a target ranking π of X, let $r := \dim \operatorname{span}\{x_i - x_j : 1 \leq i < j \leq n\}$ be the effective dimension of X. Algorithm 2 outputs a linearly realizable ranking $\widehat{\sigma}$ in polynomial time such that $d_{\mathrm{FR}}(\pi, \widehat{\sigma}) \leq 2(r+1)\,\mathrm{OPT}_{\mathrm{FR}}(X, \pi)$. Since $r \leq n-1$, this is also a $2n$-approximation.*

Proof. For every realizable σ, the first inequality in (2) gives $d_{\mathrm{KT}}(\pi, \sigma) \leq d_{\mathrm{FR}}(\pi, \sigma)$. Taking minima over the same feasible family yields $\mathrm{OPT}_{\mathrm{KT}} \leq \mathrm{OPT}_{\mathrm{FR}}$. For the output of our algorithm, the second inequality and Theorem 3.1 give

$$d_{\mathrm{FR}}(\pi, \widehat{\sigma}) \leq 2 d_{\mathrm{KT}}(\pi, \widehat{\sigma}) \leq 2(r+1)\,\mathrm{OPT}_{\mathrm{KT}} \leq 2(r+1)\,\mathrm{OPT}_{\mathrm{FR}}.$$

The running time does not change, and $r + 1 \leq n$ gives the stated dimension-independent bound. □

D Missing Details from Section 4

D.1 Why one sample estimates every weight vector

Lemma 4.2 must cover a weight vector chosen *after* sampling. Estimating agreement for one fixed weight vector would not suffice: optimizing the sample can select weights that perform better on that sample than on the full comparison set. We therefore need an estimate that holds for all weight vectors at once.

Proof of Lemma 4.2. Represent the comparison vectors in $\mathbb{R}^r$, as in Section 2. Let $F(u)$ and $F_S(u)$ denote the agreement counts in these coordinates. Each weight vector accepts the comparison vectors on the positive side of a hyperplane through the origin, a set called a *homogeneous halfspace*. Formally, weight vector u accepts $H_u = \{y \in \mathbb{R}^r : \langle u, y \rangle > 0\}$. The set's *indicator* equals 1 on accepted vectors and 0 otherwise. Sample comparisons independently and uniformly from C, with replacement. The probability that H_u accepts a sampled comparison is the fraction $F(u)/M$ over all comparisons; the fraction of accepted sample occurrences is $F_S(u)/s$. Repeated comparisons count separately in this estimate.

A family *shatters* a finite set if it can accept exactly any requested subset of that set. Its *Vapnik–Chervonenkis (VC) dimension* is the largest size of a set it shatters; this measures how many acceptance choices the family can express. Homogeneous halfspaces in $\mathbb{R}^r$ have VC dimension at most r. For the upper bound, take any $r+1$ vectors y_i. They have a nonzero linear dependence $\sum_i c_i y_i = 0$. Multiply the coefficients by -1, if needed, so some $c_i > 0$. Request acceptance of all vectors with $c_i > 0$ and rejection of those with $c_i < 0$. Any realizing u would make $\sum_i c_i \langle u, y_i \rangle > 0$: the positive-coefficient terms are positive, and the negative-coefficient terms are nonnegative. This contradicts the dependence. Coefficients equal to zero do not affect the argument. Hence the set cannot be shattered.

The standard uniform estimation theorem for a family with VC dimension at most r states that, for a universal constant C,

$$s \geq C\eta^{-2}\big(r \log(2/\eta) + \log(2/\delta)\big)$$

independent samples suffice to ensure, with probability at least $1-\delta$, that every member's sample fraction differs from its true probability by at most η [4]. The theorem applies to any probability distribution, including the uniform distribution on our finite comparison list. Substituting the halfspace family gives the event $\sup_u |F_S(u)/s - F(u)/M| \le \eta$, which is precisely (4). Weight vectors on hyperplanes cause no ambiguity: a zero inner product has indicator 0, as in the definition of F. Compression preserves all comparison signs, so the bound holds simultaneously for all original weight vectors $w \in \mathbb{R}^d$, including a vector selected after sampling. □

The conclusion holds simultaneously for a vector maximizing agreement on the sample and one maximizing agreement on all comparisons. It therefore bounds both the possible overestimate for the former vector and the possible underestimate for the latter. The main proof loses at most $2\eta M$ agreements between these two steps. Its lower bound $\mathrm{OPT}_{\mathrm{agr}} \ge M/2$ turns this additive loss into the desired multiplicative loss.

D.2 Exact optimization of the sampled comparisons

The sampling guarantee only helps if the algorithm finds the best weight vector for the sampled list. We prove Lemma 4.3 by showing that enumerating feasible subsets finds the sample optimum. We account for ties, repeated comparisons, and the number of bits needed to represent the returned weights.

Proof. The possible agreement counts form a nonempty subset of the integers $\{0, \dots, s\}$, so some weight vector attains the largest count. If it has ties, Lemma 2.1 makes the comparison system of its positive sample pairs feasible. Rational linear programming finds a rational solution, and Lemma 2.2 preserves its required comparisons while breaking all ties. The resulting weight vector therefore agrees on every originally positive sample comparison. Its count cannot decrease, so a tie-free weight vector $w_S^\star$ that maximizes the sample objective exists.

Let $T^\star = \{t \in [s] : \langle w_S^\star, \Delta_{e_t} \rangle > 0\}$. Positive scaling makes its comparison system feasible by Lemma 2.1. When enumeration reaches $T^\star$, it finds a rational solution and breaks ties using Lemma 2.2. The candidate still agrees at every position in $T^\star$, so its count is at least $F_S(w_S^\star)$. No weight vector can exceed the sample optimum. Retaining the candidate with largest count therefore finds that optimum.

There are 2^s subsets. Each iteration solves a rational linear program, performs polynomial-time tie-breaking, and evaluates s inner products. The bounds on the number of bits used in these steps give time $2^s \,\mathrm{poly}(n, d, L_X, s)$. Repeated comparisons impose the same inequality, so one constraint per distinct pair suffices for feasibility. Evaluation still counts every sampled occurrence; thus removing duplicate constraints does not change the objective being optimized. □

E Missing Details from Section 5: Hardness Proofs

We first prove the paired-point reduction from Section 5.1, then use it to establish the approximation lower bounds in Section 5. We also give a separate reduction for exact optimization in logarithmic ambient dimension. As in the main paper, zero and negative left-hand sides both violate a strict inequality. We count every repeated inequality separately.

E.1 The paired-point reduction

We prove the two directions of Lemma 5.1. First, we turn weights for the inequalities into a ranking that keeps each point pair consecutive. We then show how any realizable ranking gives weights for the inequalities, even if items from other pairs appear between the two items of a pair.

Proof. Let the input inequalities be $\langle a_\ell, z \rangle > 0$, $\ell \in [m]$, with nonzero rational $a_\ell \in \mathbb{Q}^q$. Create $p_\ell^+ = (m-\ell, a_\ell)$ and $p_\ell^- = (m-\ell, 0)$, in the target order $(p_1^+, p_1^-, \dots, p_m^+, p_m^-)$. Within a pair, the points differ because $a_\ell \neq 0$. Different pairs have different first coordinates. Thus all $2m$ points are distinct, including when coefficient vectors repeat. The coordinates have polynomial encoding length.

Constructing a ranking from inequality weights. The possible violation-counts are integers between 0 and m, so a vector z attains the smallest count. Use the perturbation described in Appendix B.1 to preserve every positive inner product and make all inner products nonzero. The perturbation cannot increase the count; since z was optimal, the resulting vector is still optimal. A rational vector with the same signs also exists: require each signed inner product to be at least 1, scale the real sign realization, and solve the resulting rational linear program. Standard rational linear-programming bounds give a solution of polynomial encoding length. This existence argument does not require knowing the optimal sign pattern algorithmically.

Let $A = \max_\ell |\langle a_\ell, z \rangle|$ and set $\lambda = 2A + 1$. Use the weight vector $w = (\lambda, z)$. For pairs $\ell < h$, any point in pair ℓ has score exceeding any point in pair h by at least $\lambda(h-\ell) - 2A \ge 1$. Thus the ranking keeps each block consecutive and places the blocks in their target order. Within block ℓ, the positive point precedes the negative point exactly when $\langle a_\ell, z \rangle > 0$. Every violation therefore makes one adjacent swap. Different blocks introduce no other reversed pairs. If the violation-count is v, this ranking has Kendall's Tau distance v and footrule distance $2v$: each swapped point moves one position. Choosing weights with the minimum violation-count gives the required upper bounds.

Obtaining inequality weights from any realizable ranking. Suppose a ranking $\sigma = (b_1, \dots, b_{2m})$ is realizable. Find a rational realizing weight vector by solving the adjacent-score constraints $\langle w, b_j - b_{j+1} \rangle \ge 1$, $j \in [2m-1]$. Their feasibility follows by scaling a strict realization; transitivity then gives the full order. The linear program and its rational solution have polynomial size. Let $w = (\lambda, z)$ denote this solution. The inner product $\langle a_\ell, z \rangle$ is nonzero because the two points receive distinct scores. If it is negative, σ reverses that pair. These are distinct unordered pairs, so their number is at most the Kendall's Tau distance.

For the footrule bound, let a reversed pair occupy target positions $t, t+1$, and output positions u, v, respectively, with $u > v$. Its two items contribute $|u-t| + |v-(t+1)| \ge |(u-v)+1| \ge 2$. The pairs have disjoint items, so we can add these lower bounds without counting any item's displacement twice. Hence twice the violation-count is at most the full footrule distance, even when items from different blocks appear between one another.

Every resulting violation-count is at least $\mathrm{OPT}_{\mathrm{viol}}$. Combining these lower bounds with the constructed upper bounds proves both distance identities. The agreement identity follows because agreement and Kendall's Tau disagreement add to $\binom{2m}{2}$. Solving the realization constraints and taking the last q coordinates of the resulting weights also proves the polynomial-time claim. □

E.2 Hardness from Learning Halfspaces

The approximation lower bounds follow from Learning Halfspaces, defined in Section 5. The reduction in [44] already gives homogeneous strict inequalities, so it has exactly the form required by Lemma 5.1. We state its guarantees in our notation before applying the paired-point construction. Theorem E.1 rules out approximating the minimum violation-count within factors that grow with the number of bits needed to describe the inequality system.

Theorem E.1. *Given nonzero rational vectors $a_1, \ldots, a_m \in \mathbb{Q}^q$, consider finding $z \in \mathbb{R}^q$ that minimizes the number of violated inequalities $\langle a_\ell, z\rangle > 0$. Let S be the binary description length of this system. There are constants $c_0, c_1 > 0$ such that this problem has no deterministic polynomial-time $S^{c_0/\log\log S}$-approximation unless* P = NP. *Under PGC, a deterministic polynomial-time S^{c_1}-approximation would also imply* P = NP.

Connection to classification and input size. Theorem 4.2 and Corollary 4.5 of [44] state the gaps for Learning Halfspaces. There, the error is the fraction of misclassified examples, the noise is the smallest possible error, and the failure ratio divides error by noise. Multiplying both fractions by the number of examples gives exactly the ratio of a violation-count to its optimum.

To express the classification requirements as inequalities, consider an example $u_\ell \in \mathbb{Q}^{q-1}$ with label $y_\ell \in \{-1, 1\}$. A halfspace with weight vector a and threshold b classifies this example correctly exactly when

$$\big\langle (y_\ell u_\ell, -y_\ell), (a, b)\big\rangle > 0.$$

Thus every labeled example is one homogeneous strict inequality, and a misclassification, including an example on the separating hyperplane, is exactly a violation. The coefficient vector is nonzero because its last coordinate is $-y_\ell$. Section 4.3 of the cited paper uses this homogeneous formulation and includes constraints that force its additional scaling variable to be positive.

The cited gaps are stated in terms of the size of the constructed Learning Halfspaces instance. Its list of rational coefficient vectors has a binary description length S bounded by a fixed polynomial in that size. Decreasing the positive exponent constants, if necessary, converts the published factors into the factors stated in Theorem E.1. This follows from the eventual monotonicity of $\log t/\log\log t$; the same elementary calculation is given below when we express the lower bound in the number of ranking items.

The conjecture used for the stronger gap. In a *projection game*, the input is a bipartite graph with a finite set of possible labels at each side. Every edge specifies a map from its left label set to its right label set. An assignment satisfies an edge when the right endpoint's label equals the image of the left endpoint's label. Its *value* is the fraction of satisfied edges. PGC asserts that, for some $c > 0$ and every $\gamma \geq N^{-c}$, Boolean satisfiability on input size s reduces efficiently to a game of size $N = s^{1+o(1)}\,\mathrm{poly}(1/\gamma)$ and label-set size $\mathrm{poly}(1/\gamma)$, distinguishing value 1 from value at most γ [43, Conjecture 2.2]. The weaker polynomial-size version suffices for the gap we use. We assume PGC in addition to P ≠ NP.

Proof of the item-dependent bound. We prove Theorem 5.2. Given an inequality system, apply the paired-point construction. If an α-approximation for Kendall's Tau distance returns cost at most $\alpha\,\mathrm{OPT}_{\mathrm{KT}}$, Lemma 5.1 gives weights with at most $\alpha\,\mathrm{OPT}_{\mathrm{viol}}$ violations. For footrule, the returned distance is at most $2\alpha\,\mathrm{OPT}_{\mathrm{viol}}$, and the same lemma gives weights whose violation-count is at most half the returned distance. Thus either ranking approximation gives an inequality approximation with the same factor.

The paired-point instance has $n = 2m = O(S)$ items because the binary description of the inequality system contains all m inequalities. In particular, $n \leq S^2$ for all sufficiently large instances. The function $\log t/\log\log t$ is eventually increasing. Consequently,

$$\log\big(n^{c/\log\log n}\big) \leq \frac{2c\log S}{\log\log S}.$$

Choose $c < c_0/2$. The resulting approximation factor is smaller than the lower bound in Theorem E.1. Similarly, choosing $0 < c' < c_1/2$ contradicts the PGC gap because $n^{c'} \leq S^{2c'} < S^{c_1}$. These factor expressions are asymptotic; changing their values for the finitely many smaller item counts does not affect the hardness statement.

Proof of the dimension-dependent bound. We prove Theorem 5.3. Start with the paired-point data in $\mathbb{Q}^{q+1}$. Choose a rational basis matrix P of the span of all differences, and compute coordinates y_i with $x_i - x_1 = Py_i$. The matrix has full column rank r. Every original weight w gives compressed weight $u = P^\top w$, with identical pairwise score differences. Conversely, for every $u \in \mathbb{R}^r$, the system $P^\top w = u$ has a solution because $P^\top$ has rank r. Thus compression preserves the whole family of rankings induced without ties and all three optimum values. It preserves distinctness as well: equal y_i, y_j would imply equal x_i, x_j. Gaussian elimination constructs the rational coordinates with polynomial encoding length.

The compressed differences span $\mathbb{R}^r$, so its ambient and effective dimensions coincide. Rename this ambient dimension as d. It satisfies $d = r \leq q + 1 \leq S + 1$. Given any returned ranking, solve its realization constraints on the original paired points and obtain inequality weights as in Lemma 5.1. We do not need the approximation algorithm to return its weights.

For sufficiently large S, use $d \leq S + 1$ and eventual monotonicity to bound $d^{a/\log\log d}$ by $S^{c_0/\log\log S}$, choosing $a > 0$ sufficiently small. For PGC, choose $0 < b < c_1$; then $(S+1)^b \leq S^{c_1}$ for sufficiently large S. Therefore either such dimension-dependent algorithm would give an approximation excluded by Theorem E.1. This proves the theorem with $d = r$, without relying on redundant zero coordinates. As above, the factor expressions are interpreted asymptotically; the finitely many dimensions outside the monotonic range do not affect the hardness statement.

This argument does not force $d = O(\log n)$, nor does it prove hardness for every factor $o(d)$. For example, $d/\log d$ is larger than both the stated $d^{a/\log\log d}$ factor and a sufficiently small-power d^b factor for large d. The reduction therefore leaves the gap to the main algorithm's $O(r)$ guarantee open.

E.3 Exact hardness with logarithmic ambient dimension

We next show that logarithmic dimension does not admit deterministic polynomial-time exact optimization or a deterministic FPTAS under the clique assumption stated below. In k-Clique, the input is a graph G and an integer k, and we ask whether k distinct vertices are pairwise adjacent. Under FPT $\neq$ W[1], no deterministic algorithm solves this problem in time $f(k)\,\mathrm{poly}(|G|)$, where $|G|$ is the number of bits needed to describe the graph [22].

We turn the choice of k pairwise adjacent vertices into linear inequalities, then into a ranking instance whose optimum determines whether the clique exists. The construction takes $f(k)\,\mathrm{poly}(|G|)$ time and ensures $d = \Theta(\log n)$. A deterministic polynomial-time exact ranking algorithm would therefore give the clique algorithm ruled out above. A deterministic FPTAS would also return the exact optimum by setting $\varepsilon = 1/(2n^2)$, because every objective value is an integer smaller than n^2. Excluding algorithms that may err with small probability requires assuming that no such clique algorithm runs in $f(k)\,\mathrm{poly}(|G|)$ time.

Theorem E.2. *Consider instances consisting of $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π of X. Unless* FPT = W[1]*, no deterministic polynomial-time exact algorithm or deterministic FPTAS exists for minimizing Kendall's Tau distance, minimizing Spearman footrule distance, or maximizing Kendall agreement over rankings induced by linear scores, even when $d = O(\log n)$. We may require ambient dimension $d = \Theta(\log n)$.*

Preparing the graph. We may assume $k \geq 2$; smaller cases are immediate. Add isolated vertices until the graph has $N \geq \max\{2^k, k\}$ vertices, numbered $1, \ldots, N$. An isolated vertex cannot enter a clique of size at least two, so the answer does not change. The resulting graph has at most $2^k + k + |G|$ vertices, and adding these vertices takes $f(k)\,\mathrm{poly}(|G|)$ time. In particular, $k \leq \log N$.

Choosing vertices and checking edges. Use variables $u_1, \ldots, u_k$ to choose vertex numbers, and put $\tau = 1/[4(N+2)]$. An *affine expression* is a linear expression plus a constant. For each position i and vertex a, let $h_{i,a} = u_i - a$. For each position pair $i < j$ and each ordered edge (a, b), introduce $h_{ij,ab} = u_i - a + N(u_j - b)$. Use both orientations of every undirected edge. For each expression h, include the two strict inequalities $\tau + h > 0$ and $\tau - h > 0$. There are $Q = kN + \binom{k}{2}\,2|E|$ such pairs.

Each pair has at least one satisfied inequality, and both are satisfied exactly when $|h| < \tau$. In a vertex-choice group i, at most one pair can have both inequalities satisfied because different integers a are separated by at least one and $2\tau < 1$. Likewise, in an edge group i, j, at most one pair can have both inequalities satisfied: the integers $a + Nb$, for $a, b \in [N]$, are distinct, and hence at least one apart. To check distinctness, equality would give $a - a' = N(b' - b)$; its left side has absolute value at most $N - 1$, so both differences must be zero. There are $b_k = k + \binom{k}{2}$ groups in total.

Lemma E.3 identifies when the maximum possible number of inequalities is satisfied, thereby turning the real-valued choices into a clique decision.

Lemma E.3. *No assignment satisfies more than $Q + b_k$ inequalities in the affine system. An assignment attains this value if and only if the graph contains a k-clique.*

Proof. The bound follows because each pair contributes one guaranteed satisfied inequality and each group contributes at most one additional satisfied inequality. If vertices $a_1, \ldots, a_k$ form a clique, choose $u_i = a_i$. Each vertex-choice group has a zero expression, and each edge group contains the ordered edge (a_i, a_j) with zero expression. Consequently, each of the b_k groups contributes one additional satisfied inequality.

Conversely, attaining the bound requires some pair in every group to have both inequalities satisfied. Let a_i be the vertex associated with that pair in group i, so $|u_i - a_i| < \tau$. For each $i < j$, let (a, b) be the ordered edge whose group expression has absolute value less than τ. The integer $a_i - a + N(a_j - b)$ has absolute value less than $(N+2)\tau = 1/4$, using the two vertex-choice bounds and the edge bound. The integer must therefore be zero. The equality $a_i - a = N(b - a_j)$, together with $|a_i - a| \leq N - 1$, forces $a_i = a$ and $a_j = b$. Thus the chosen vertices have an edge between every pair. The graph has no self-loops, so they are distinct and form a clique. □

Making every inequality homogeneous. Introduce one additional variable t. Replace every constant by that constant times t: for vertex groups, use $h = u_i - at$, and for edge groups use $h = u_i + Nu_j - (a + Nb)t$. Replace each pair by $\tau t + h > 0$ and $\tau t - h > 0$, and add the inequality $t > 0$. The resulting system has $q = k + 1$ variables and $m = 2Q + 1$ inequalities, all with nonzero rational coefficient vectors. Lemma E.4 shows that the additional inequality prevents the new variable from changing the intended threshold.

Lemma E.4. *The homogeneous system has at most $B = Q - b_k$ violations if and only if the graph contains a k-clique. If the graph contains a k-clique, the optimum is exactly B; otherwise, it is at least $B + 1$.*

Proof. If $t > 0$, divide all variables by t. The first $2Q$ inequalities become the affine system in Lemma E.3, and the additional inequality is satisfied. Their maximum possible satisfaction count is $Q + b_k + 1$, giving at least $(2Q + 1) - (Q + b_k + 1) = B$ violations. Equality occurs exactly when there is a clique.

If $t \leq 0$, the two left sides in each pair sum to $2\tau t \leq 0$, so they cannot both be positive. At most Q inequalities are satisfied, and $t > 0$ also fails. There are then at least $Q + 1 > B$ violations. This excludes all unintended values of t. Finally, $Q \geq kN \geq b_k$, so B is nonnegative. The objectives are integers, yielding the claimed one-unit gap. □

Transferring the threshold to rankings. Apply the paired-point reduction. It creates $n = 4Q + 2$ distinct points in ambient dimension $d = k + 2$. The clique answer is yes exactly when $\mathrm{OPT}_{\mathrm{KT}} \leq B$, equivalently $\mathrm{OPT}_{\mathrm{FR}} \leq 2B$, equivalently $\mathrm{OPT}_{\mathrm{agr}} \geq \binom{n}{2} - B$. The coefficients have polynomial encoding length in k and $\log N$, and $Q = O(k^2 N^2)$. Together with the padding, the entire construction takes $f(k)\,\mathrm{poly}(|G|)$ time and output size. Moreover, $d = k + 2 \leq \log N + 2 = O(\log n)$. A polynomial-time exact algorithm for any of the ranking objectives under this dimension restriction would therefore solve k-Clique in the excluded parameterized time.

To require $d = \Theta(\log n)$, append $\lceil \log n \rceil$ zero coordinates to every point. No score difference or realizable ranking changes. The new ambient dimension lies between $\log n$ and $O(\log n)$, and

the output size remains bounded by $f(k)\,\mathrm{poly}(|G|)$. This padding controls ambient dimension only; it does not assert that the effective dimension is $\Theta(\log n)$.

Why an FPTAS would give exact optimization. Suppose an FPTAS exists for one of the restricted objectives. Run it with $\varepsilon = 1/(2n^2)$. Every feasible Kendall's Tau distance and agreement value is at most $\binom{n}{2} < n^2$, and every footrule distance is at most $n(n-1) < n^2$. For minimization, the returned cost exceeds optimum by less than $\varepsilon n^2 = 1/2$. For maximization, the optimum minus the returned value is less than $1/2$. The algorithm returns a realizable ranking with integer objective, so either guarantee forces exact optimality. Its running time is polynomial in the input size and $1/\varepsilon = 2n^2$, hence polynomial overall, contradicting the exact lower bound. For randomized schemes this argument rules out schemes with an always-correct guarantee; excluding bounded-error randomized exact algorithms (correct with probability at least $2/3$) requires the corresponding randomized clique assumption. Our exact and FPTAS hardness statements use the standard deterministic convention.

F Additional Hardness under Alternative Assumptions

The paired-point construction also transfers classical lower bounds for minimizing the number of unsatisfied linear inequalities. These bounds are weaker than those in Section 5, but give a constant-factor lower bound under $\mathsf{P} \neq \mathsf{NP}$ and a logarithmic lower bound under a different complexity assumption. Both concern arbitrary dimension, not logarithmic dimension.

The *minimum unsatisfied homogeneous strict linear relations* problem asks for a vector that violates as few input inequalities as possible. More specifically, an instance of $\textsc{Min-ULR}^{>}_{\mathrm{hom}}$ consists of nonzero $a_1, \ldots, a_m \in \{-1, 0, 1\}^q$ and minimizes the number of violations of $\langle a_\ell, z\rangle > 0$. Amaldi and Kann prove the following approximation lower bounds [3]. The condition $m = 2q$ lets us express the logarithmic factor in the number of ranking items without losing its leading constant.

Theorem F.1. *Given nonzero vectors $a_1, \ldots, a_m \in \{-1, 0, 1\}^q$, let $\textsc{Min-ULR}^{>}_{\mathrm{hom}}$ minimize the number of violated inequalities $\langle a_\ell, z\rangle > 0$ over $z \in \mathbb{R}^q$. This problem has no deterministic polynomial-time constant-factor approximation unless $\mathsf{P} = \mathsf{NP}$. For every fixed $0 < \varepsilon < 1$, it has no deterministic polynomial-time $(1-\varepsilon)\ln q$-approximation unless $\mathsf{NP} \subseteq \mathsf{DTIME}(q^{O(\log\log q)})$. The hard instances use $m = 2q$.*

Deriving the inequality system from a graph. The reduction starts from a graph with q vertices. A *dominating set* is a vertex subset such that every vertex is in the subset or has a neighbor in it. The *closed neighborhood* of a vertex consists of that vertex and its neighbors; let $N[i]$ denote this set for vertex i. For each vertex i, use the two inequalities $-z_i > 0$ and $\sum_{j\in N[i]} z_j > 0$. These are $2q$ homogeneous strict inequalities, each with a nonzero coefficient vector whose entries belong to $\{-1, 0, 1\}$.

If D is a dominating set, assign $z_i = q$ for $i \in D$ and $z_i = -1$ otherwise. Every neighborhood sum is positive, because it contains a selected vertex and at most $q-1$ negative terms. Exactly $|D|$ inequalities of the first kind fail. Conversely, given any vector, let $D_0 = \{i : z_i \geq 0\}$ and add to it every vertex whose closed neighborhood misses D_0. Each added vertex has an entirely negative neighborhood, so its sum inequality fails. Each member of D_0 fails its first inequality. The resulting set dominates the graph and its size is at most the violation-count. This proves equality of the optima and preserves every approximation factor. The dominating-set lower bounds used by Amaldi and Kann therefore give the stated theorem. Here ln denotes the natural logarithm, and $\mathsf{DTIME}(g(q))$ denotes decision problems solvable by a deterministic algorithm in time $g(q)$. The stronger complexity assumption excludes time of the form $q^{O(\log\log q)}$ for all NP problems, not merely ordinary polynomial time.

We now apply Lemma 5.1 to this inequality system. Theorem F.2 states the resulting bounds under these alternative assumptions, separately from the stronger Learning Halfspaces lower bounds.

Theorem F.2. *Consider pairwise distinct vectors $x_1, \ldots, x_n \in \mathbb{Q}^d$, let $X = \{x_1, \ldots, x_n\}$, and let π be a target ranking of X. The dimension d is part of the input. Unless $\mathsf{P} = \mathsf{NP}$, the minimum-disagreement problems under Kendall's Tau and Spearman footrule distances admit no deterministic polynomial-time constant-factor approximation. For every fixed $0 < \varepsilon < 1$, neither admits a deterministic polynomial-time $(1-\varepsilon)\ln n$-approximation unless $\mathsf{NP} \subseteq \mathsf{DTIME}(n^{O(\log\log n)})$. These bounds hold with $d = \Theta(n)$.*

Proof. Apply the paired-point construction to the homogeneous strict inequality system. A constant-factor Kendall's Tau approximation would give weights with the same approximation factor for the violation-count. For footrule, the optimum is twice the minimum violation-count, and Lemma 5.1 gives weights with at most half as many violations as the returned distance, preserving the factor again. Both conversions take polynomial time, contradicting Theorem F.1. The inequality system has $m = 2q$, so the ranking instance has $n = 4q$ and $d = q + 1 = \Theta(n)$.

For the logarithmic factor, fix ε and use the inequality bound with $\varepsilon/2$. For sufficiently large q, $(1-\varepsilon)\ln(4q) \leq (1-\varepsilon/2)\ln q$, because the additive term $(1-\varepsilon)\ln 4$ is eventually smaller than $(\varepsilon/2)\ln q$. The resulting approximation factor would therefore contradict Theorem F.1. Replacing q by $n = 4q$ also preserves the stated time class, which proves the second claim. □

G Missing Details from Section 1.1: Polynomial Scores

Polynomial scores can capture interactions between attributes that a linear score cannot express. For example, a term $x_1^2 x_2$ lets the contribution of one attribute depend on another. By treating monomials as additional features, we extend both the maximum-agreement scheme from Section 4 and the minimum-disagreement approximations from Section 3 to every fixed degree. The approximation guarantees depend on the effective dimension of these expanded features. We first define this representation and prove the approximation guarantees, including the quadratic case. We then give a separate reduction that establishes minimum-disagreement hardness for every fixed degree.

G.1 Polynomial scores and their feature representation

Fix a positive integer p, the maximum degree allowed; it is a constant, not part of the input. A *monomial* multiplies feature coordinates raised to nonnegative integer powers, and its *total degree* is the sum of those powers. More specifically, for an exponent tuple $\alpha = (\alpha_1, \ldots, \alpha_d) \in \mathbb{Z}^d_{\geq 0}$, let $x^\alpha = \prod_{j=1}^d x_j^{\alpha_j}$ and $|\alpha| = \sum_{j=1}^d \alpha_j$. A factor with exponent zero equals 1, including when its coordinate is zero. Let $\mathcal{I}_p = \{\alpha : 1 \leq |\alpha| \leq p\}$. An *unrestricted polynomial score of degree at most p* combines these monomials with arbitrary real coefficients, including negative ones. Since rankings depend only on score differences, we omit the constant monomial and let

$$f_a(x) = \sum_{\alpha \in \mathcal{I}_p} a_\alpha x^\alpha.$$

The *monomial feature map* lists every nonconstant monomial of degree at most p. Formally, let $\Phi_p(x) = (x^\alpha)_{\alpha \in \mathcal{I}_p}$, using the same order of exponent tuples for all items. Its number of coordinates is $D_p = \binom{d+p}{p} - 1$. The *expanded effective dimension*, denoted by r_p, is the dimension spanned by the differences $\Phi_p(x_i) - \Phi_p(x_j)$, over $i < j$. Thus $r_p \leq \min\{D_p, n-1\}$. We optimize over the rankings induced without ties by these polynomial scores. Let $\mathrm{OPT}^{(p)}_{\mathrm{agr}}$ denote their largest Kendall agreement with π, and let $\mathrm{OPT}^{(p)}_{\mathrm{KT}}$ and $\mathrm{OPT}^{(p)}_{\mathrm{FR}}$ denote their smallest Kendall's Tau and Spearman footrule distances from π, respectively.

Table 8 summarizes the polynomial-scoring approximation guarantees and hardness bounds.

G.2 Approximating maximum agreement

Applying the linear agreement scheme to the expanded features gives the same approximation ratio and success probability for polynomial scores. Theorem G.1 gives this guarantee and includes the cost of computing the monomials and representing them as rational numbers. This cost is polynomial for every fixed degree. Let poly_p denote a polynomial whose constants and exponent may depend on p, but not on the effective dimension or the accuracy parameters. A subscript p on O_p likewise allows the hidden constant to depend on the fixed degree.

Theorem G.1. *Fix an integer $p \geq 1$, and let $n \geq 2$. For pairwise distinct vectors $x_1, \ldots, x_n \in \mathbb{Q}^d$, let $X = \{x_1, \ldots, x_n\}$, and let π be a target ranking of X. Let r_p be the dimension of the span of the vectors $\Phi_p(x_i) - \Phi_p(x_j)$ over all $1 \leq i < j \leq n$. There exists a randomized algorithm that, given X, π, and $0 < \varepsilon, \delta < 1$, outputs a polynomial score of total degree at most p with rational coefficients and pairwise distinct item scores, and runs in time*

$$2^{O((r_p \log(2/\varepsilon) + \log(2/\delta))/\varepsilon^2)} \, \mathrm{poly}_p(n, d, L_X).$$

With probability at least $1 - \delta$, its induced ranking $\widehat{\sigma}$ satisfies $\mathrm{agr}_{\mathrm{KT}}(\pi, \widehat{\sigma}) \geq (1 - \varepsilon)\,\mathrm{OPT}^{(p)}_{\mathrm{agr}}(X, \pi)$. For fixed p, the scheme is FPT jointly in r_p, $1/\varepsilon$, and $\log(1/\delta)$. For fixed failure probability, it is a randomized PTAS when $r_p = O(\log n)$; the original dimension bound $d = O((\log n)^{1/p})$ suffices. If $d = O(\log n)$, then for fixed p, ε, δ its running time is $2^{O_p((\log n)^p)} \, \mathrm{poly}_p(n, L_X)$.

The algorithm has three steps. First, enumerate the exponent tuples and compute $\Phi_p(x_i)$ for every item. Next, run Algorithm 3 on the expanded vectors with the original target order and the requested accuracy and failure probability. Finally, use its returned weight on each expanded feature as the coefficient of the corresponding monomial. The resulting polynomial orders the original items exactly as the returned linear score orders their expanded vectors.

Proof. We first bound the size of the expanded instance. To count all monomials of degree at most p, including the constant, assign each exponent tuple an extra entry $p - |\alpha|$. The resulting $d+1$ nonnegative entries sum to p. An arrangement of p identical degree units and d separators specifies exactly one such tuple: the separators divide the units among the $d+1$ entries. Choosing their positions gives $\binom{d+p}{p}$ tuples. Removing the constant monomial gives the claimed D_p. For fixed p, this is $O_p(d^p)$. We can enumerate the tuples directly: for each degree $t \in \{1, \ldots, p\}$, list the nondecreasing sequences of t feature indices from $[d]$. Counting the occurrences of each index gives exactly one exponent tuple. Every monomial appears once, so this enumeration requires polynomial work per tuple.

Each expanded coordinate multiplies at most p input coordinates, counting repetitions. If an input rational has at most L_X bits, the numerator and denominator of such a product have $O(p(L_X + 1))$ bits before reduction; reducing the fraction cannot increase this bound. We compute these products exactly by rational arithmetic. Enumerating the exponent tuples and computing all nD_p coordinates therefore takes $\mathrm{poly}_p(n, d, L_X)$ time. The maximum encoding length of one expanded coordinate, denoted by L_{Φ_p}, is also bounded by such a polynomial.

We next prove that expansion preserves the feasible rankings. Every polynomial coefficient vector a gives the linear score $\langle a, \Phi_p(x) \rangle = f_a(x)$. Conversely, every linear score on the expanded coordinates gives a polynomial of degree at most p on the original features. In particular, for every pair of items,

$$f_a(x_i) - f_a(x_j) = \langle a, \Phi_p(x_i) - \Phi_p(x_j) \rangle.$$

The two representations give identical score differences, so they produce the same ties and the same rankings induced without ties. Since $p \geq 1$, the expanded vectors include every original coordinate, so distinct original items remain distinct. We can therefore apply the linear algorithm. This equality of the ranking families also shows that the optimal linear agreement on the expanded data equals $\mathrm{OPT}^{(p)}_{\mathrm{agr}}$.

Now run Algorithm 3 on the expanded instance. Theorem 4.4 applies with ambient dimension D_p, effective dimension r_p, and encoding length L_{Φ_p}. Its tie-breaking step gives a rational coefficient vector $\widehat{a}$ with distinct scores on all expanded items. The polynomial with these coefficients gives the same distinct scores and hence the same ranking $\widehat{\sigma}$ on the original items. With probability at least $1 - \delta$, its agreement is at least $1 - \varepsilon$ times the expanded linear optimum. Equality of the two optima proves the claimed approximation guarantee.

The linear algorithm has the exponential factor stated in the theorem and a remaining factor $\mathrm{poly}(n, D_p, L_{\Phi_p})$. Our bounds on D_p and L_{Φ_p} turn its polynomial factor, together with the cost of feature expansion, into $\mathrm{poly}_p(n, d, L_X)$. This proves the stated running time and the FPT claim for each fixed p.

Table 8: Polynomial-scoring results for Kendall's Tau disagreement and agreement, for any fixed maximum degree $p \geq 1$. All approximation algorithms run in polynomial time; the PTAS is randomized. Here $r_p \leq \binom{d+p}{p} - 1$ is the expanded effective dimension. Hardness exponents may depend on p, and dimension-dependent bounds also hold with r_p in place of d.

Objective	Setting	Guarantee or lower bound	Condition
Min disagreement	arbitrary r_p	$(r_p + 1)$-approximation (Theorem G.2)	–
Max agreement	$r_p = O(\log n)$	PTAS (Theorem G.1)	–
Min disagreement	arbitrary dimension	no $n^{c_p/\log\log n}$- or $d^{a_p/\log\log d}$-approximation (Theorems G.7 and G.8)	P ≠ NP
Min disagreement	arbitrary dimension	no $n^{c'_p}$- or d^{b_p}-approximation (Theorems G.7 and G.8)	PGC, P ≠ NP
Both objectives	arbitrary dimension	no deterministic exact algorithm or FPTAS (Corollary G.9)	P ≠ NP

If $r_p = O(\log n)$, the exponential factor is a fixed power of n for fixed ε, δ, so the scheme is a PTAS. Since $r_p \leq D_p = O_p(d^p)$, the bound $d = O((\log n)^{1/p})$ ensures this condition. If instead $d = O(\log n)$, then $r_p = O_p((\log n)^p)$. Substituting into the running time gives the last claim. For $p \geq 2$, this bound need not be polynomial: the exponent on n may grow as $(\log n)^{p-1}$. □

The polynomial running-time bound requires a fixed degree. If p grows with the input, the number D_p of monomials can grow faster than any polynomial in the original input size. The equality of the ranking families still holds, but the proof does not give polynomial-time feature expansion or an FPT guarantee jointly in the degree and the other parameters.

G.3 Approximating minimum disagreement

The same representation lets us approximate the number of reversed target pairs. Unlike the agreement scheme, the following algorithm needs neither random sampling nor a bound on the effective dimension to run in polynomial time. Increasing the degree can reduce the smallest achievable disagreement, but it can also increase the approximation factor through r_p.

Theorem G.2. *Fix an integer $p \geq 1$, and let $n \geq 2$. For pairwise distinct vectors $x_1, \ldots, x_n \in \mathbb{Q}^d$, let $X = \{x_1, \ldots, x_n\}$, and let π be a target ranking of X. Let r_p be the dimension of the span of the vectors $\Phi_p(x_i) - \Phi_p(x_j)$ over all $1 \leq i < j \leq n$, and let $D_p := \binom{d+p}{p} - 1$ be the number of expanded features. There exists a deterministic algorithm that, given X and π, outputs a polynomial score of total degree at most p with rational coefficients and pairwise distinct item scores, and runs in time $\mathrm{poly}_p(n, d, L_X)$. Its induced ranking $\widehat{\sigma}$ satisfies $d_{\mathrm{KT}}(\pi, \widehat{\sigma}) \leq (r_p + 1)\,\mathrm{OPT}_{\mathrm{KT}}^{(p)}(X, \pi)$. The factor is at most $D_p + 1 = \binom{d+p}{p}$ and at most n.*

To implement the algorithm, compute the expanded vectors $\Phi_p(x_i)$ and run Algorithm 2 on them in the original target order. Each comparison now requires the weighted sum of monomial differences to be positive; the weights themselves remain unrestricted. Once Algorithm 2 finds a feasible subset of the target comparisons, it computes rational weights and breaks ties while preserving that subset. Use each resulting weight as the coefficient of its monomial. Sorting the polynomial scores gives the output. Thus the algorithm works in the expanded feature space and does not require a separate search over polynomial functions.

Proof. The construction in the proof of Theorem G.1 computes all expanded vectors in $\mathrm{poly}_p(n, d, L_X)$ time. Their ambient dimension is D_p, their effective dimension is r_p, and their maximum coordinate encoding length L_{Φ_p} is polynomial in the original input for fixed p. They remain pairwise distinct because their degree-one coordinates include all the original features.

For every coefficient vector a, the identity $f_a(x_i) - f_a(x_j) = \langle a, \Phi_p(x_i) - \Phi_p(x_j)\rangle$ holds for every pair. Conversely, every linear weight vector on the expanded coordinates defines an allowed polynomial score. Hence the linear rankings of the expanded items and the polynomial rankings of the original items form exactly the same family of orders. The target order also stays the same, so each corresponding ranking has the same Kendall's Tau distance. In particular, the minimum linear distance on the expanded instance equals $\mathrm{OPT}_{\mathrm{KT}}^{(p)}$.

Apply Algorithm 2 to this expanded instance. Theorem 3.1 applies with effective dimension r_p and gives the claimed factor. More explicitly, Lemma 3.3 bounds each removed conflict by $r_p + 1$ comparisons. Lemma 3.5 bounds the number T of these disjoint conflicts by the optimal deletion cost, which equals $\mathrm{OPT}_{\mathrm{KT}}^{(p)}$ by Lemma 3.2 and the equality of the ranking families. If D is the union of the removed conflicts, then

$$d_{\mathrm{KT}}(\pi, \widehat{\sigma}) \leq |D| \leq (r_p + 1)T \leq (r_p + 1)\,\mathrm{OPT}_{\mathrm{KT}}^{(p)}.$$

The first inequality holds because tie-breaking preserves all comparisons outside D. Use the final rational weights as polynomial coefficients. The resulting polynomial gives the same distinct scores and therefore the same pairwise orders.

The linear algorithm uses a polynomial number of rational linear-programming tests and a polynomial-time tie-breaking step. Its running time on the expanded input is therefore $\mathrm{poly}(n, D_p, L_{\Phi_p})$. The bounds on the expansion turn this into $\mathrm{poly}_p(n, d, L_X)$, including the cost of computing the monomials and outputting the rational coefficients. There is no restriction on d or r_p here: only p is fixed. Finally, $r_p \leq \min\{D_p, n - 1\}$ gives both stated factor bounds. □

The same output also approximates the minimum Spearman footrule distance. The ranking that minimizes footrule need not minimize Kendall's Tau distance, but the Diaconis–Graham inequalities relate the two optima over the same polynomial scoring family. Corollary G.3 gives the resulting guarantee without an additional optimization step.

Corollary G.3. *For every fixed integer $p \geq 1$, there exists a deterministic algorithm that, given $n \geq 2$ pairwise distinct vectors $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π, outputs a polynomial score whose induced ranking $\widehat{\sigma}$ satisfies $d_{\mathrm{FR}}(\pi, \widehat{\sigma}) \leq 2(r_p+1)\,\mathrm{OPT}_{\mathrm{FR}}^{(p)}(X, \pi)$, where*

$$r_p := \dim \operatorname{span}\{\Phi_p(x_i) - \Phi_p(x_j) : 1 \leq i < j \leq n\}.$$

The algorithm runs in time $\mathrm{poly}_p(n, d, L_X)$. Its approximation factor is also at most $2\binom{d+p}{p}$ and at most $2n$.

Proof. Every ranking induced without ties by a polynomial score has Kendall's Tau distance from π at most its footrule distance, by (2). Taking minima over the same polynomial ranking family yields $\mathrm{OPT}_{\mathrm{KT}}^{(p)} \leq \mathrm{OPT}_{\mathrm{FR}}^{(p)}$. For the output of Theorem G.2, the other Diaconis–Graham inequality gives

$$\begin{aligned} d_{\mathrm{FR}}(\pi, \widehat{\sigma}) &\leq 2 d_{\mathrm{KT}}(\pi, \widehat{\sigma}) \\ &\leq 2(r_p + 1)\,\mathrm{OPT}_{\mathrm{KT}}^{(p)} \\ &\leq 2(r_p + 1)\,\mathrm{OPT}_{\mathrm{FR}}^{(p)}. \end{aligned}$$

We reuse the same coefficients, tie-breaking, and ranking, so the algorithm remains deterministic and its running time does not change. The bounds $r_p + 1 \leq D_p + 1$ and $r_p + 1 \leq n$ give the remaining claims. □

G.4 Quadratic scores as a special case

For $p = 2$, we have $D_2 = d(d+3)/2$: the expanded vector contains the original features, their squares, and the products of distinct feature pairs. Theorem G.2 and Corollary G.3 give deterministic $(r_2 + 1)$- and $2(r_2 + 1)$-approximations for the two disagreement distances. Their factors are at most $d(d+3)/2 + 1$ and $d(d+3) + 2$, respectively, and also at most n and $2n$. Corollary G.4 gives the maximum-agreement guarantee for quadratic scores and identifies when its running time is polynomial.

Corollary G.4. *Let $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ be a set of pairwise distinct vectors, let π be a target ranking of X, and let*

$$r_2 := \dim \operatorname{span}\{\Phi_2(x_i) - \Phi_2(x_j) : 1 \leq i < j \leq n\}.$$

There exists a randomized algorithm that, given X, π, and $0 < \varepsilon, \delta < 1$, outputs a ranking $\widehat{\sigma}$ induced without ties by a quadratic score and runs in time

$$2^{O((r_2 \log(2/\varepsilon) + \log(2/\delta))/\varepsilon^2)}\,\mathrm{poly}(n, d, L_X).$$

With probability at least $1 - \delta$, $\widehat{\sigma}$ has Kendall agreement at least $1 - \varepsilon$ times the largest agreement of any quadratic score on X. It is a PTAS when $r_2 = O(\log n)$, in particular when $d = O(\sqrt{\log n})$. For $d = O(\log n)$ and fixed ε, δ, its running time is $2^{O(\log^2 n)}\,\mathrm{poly}(n, L_X)$.

Proof. Apply Theorem G.1 with $p = 2$. The monomials of degrees one and two give exactly the unrestricted quadratic family. The original dimension threshold becomes $d = O((\log n)^{1/2})$, and the logarithmic-dimension running time becomes $2^{O(\log^2 n)}\,\mathrm{poly}(n, L_X)$. These ambient-dimension conditions suffice for the stated bounds; the algorithm's actual dependence is on r_2, which can be smaller than D_2. □

G.5 Hardness for every fixed degree

Feature expansion alone does not extend the linear hardness results. A polynomial score can realize additional orders and may therefore achieve a smaller distance on an instance that is hard for linear scores. We construct item blocks that let us obtain weights for the original inequalities from *any* ranking induced by a polynomial score. The resulting approximation factor increases by at most a constant that depends on the fixed degree. We can absorb this constant by decreasing the exponents in Theorem E.1.

G.5.1 Extracting a linear coefficient. We add one coordinate, denoted by t, and choose p fixed *evaluation values* for it. A weighted sum of the scores at these values isolates the coefficient of t^{p-1}. Since the total degree is at most p, this coefficient consists of a constant and linear terms in the remaining coordinates. Evaluating at opposite feature vectors and taking the difference will cancel all but the inner product needed for the original inequality.

For $i \in \{0, \ldots, p-1\}$, set $t_i = i - (p-1)/2$ and $c_i = (\prod_{j \neq i}(t_i - t_j))^{-1}$. The evaluation values are distinct rationals and sum to zero. When $p = 1$, the empty product equals 1, so $t_0 = 0$ and $c_0 = 1$. For a polynomial $f(\rho, x, t)$, where $\rho \in \mathbb{R}$ and $x \in \mathbb{R}^q$, define the evaluation sum $\mathcal{E}_p f(\rho, x) = \sum_{i=0}^{p-1} c_i f(\rho, x, t_i)$. Lemma G.5 shows that taking the difference of these sums at opposite feature vectors gives a linear inner product. We will use it to turn ranking comparisons into inequalities on the extracted coefficient vector.

Lemma G.5. *For every polynomial $f(\rho, x, t)$ of total degree at most p, there are coefficients $\beta_0, \beta_\rho \in \mathbb{R}$ and $b \in \mathbb{R}^q$ such that $\mathcal{E}_p f(\rho, x) = \beta_0 + \beta_\rho \rho + \langle b, x \rangle$. Consequently,*

$$\mathcal{E}_p f(\rho, a) - \mathcal{E}_p f(\rho, -a) = 2\langle b, a \rangle \qquad \text{for every } a \in \mathbb{R}^q.$$

These coefficients are rational when f has rational coefficients.

Proof. Let $L_i(t) = \prod_{j \neq i}(t - t_j)/(t_i - t_j)$. This polynomial equals 1 at t_i, equals 0 at every other evaluation value, and has leading coefficient c_i. For $0 \leq k \leq p-1$, the interpolation identity $t^k = \sum_i t_i^k L_i(t)$ holds: both sides have degree at most $p-1$ and agree at the p distinct evaluation values, so their difference must be zero. Comparing coefficients of t^{p-1} gives $\sum_i c_i t_i^k = 0$ for $k < p-1$, and gives 1 when $k = p-1$.

We must also eliminate the possible degree-p term. The polynomial $t^p - \prod_i (t - t_i)$ has degree at most $p-1$ and equals t_i^p at each evaluation value. Its coefficient of t^{p-1} is $\sum_i t_i = 0$, so the same interpolation argument gives $\sum_i c_i t_i^p = 0$. This argument also covers $p = 1$.

Collecting powers of t gives $f(\rho, x, t) = \sum_{k=0}^{p} f_k(\rho, x) t^k$. The identities just proved give $\mathcal{E}_p f(\rho, x) = f_{p-1}(\rho, x)$. The total-degree bound implies that f_{p-1} has degree at most one in (ρ, x), so it consists of a constant and linear terms. Evaluating at a and $-a$ cancels its constant and its ρ term. For rational f, its coefficients, including b, are rational as well. □

G.5.2 Encoding one inequality in an item block. Consider the homogeneous strict inequality system $\mathcal{I} = (\langle a_\ell, z \rangle > 0 : \ell \in [m])$ from Section 5.1, with nonzero rational $a_\ell \in \mathbb{Q}^q$. Let S be the number of bits needed to describe this system. As before, repeated inequalities count separately. Choose $v_i = i + 1$ when $c_i > 0$, and $v_i = -(i+1)$

otherwise. Let g be the rational polynomial of degree at most $p-1$ that satisfies $g(t_i) = v_i$; explicitly, $g(t) = \sum_i v_i L_i(t)$.

For each inequality, create the $2p$ items

$$u_{\ell,\eta,i} = (m-\ell, \eta a_\ell, t_i), \qquad \eta \in \{-1, 1\}, \quad 0 \le i < p.$$

Let π be the target ranking that lists blocks in increasing ℓ. Inside a block, it lists the items in decreasing order of ηv_i. The values v_i have distinct absolute values, so the $2p$ numbers ηv_i are distinct. The coefficient ηc_i has the same sign as ηv_i. Thus every item with a positive coefficient in the extraction sum precedes every item with a negative coefficient. We will show that any polynomial score that preserves this block order must give a positive inner product for the corresponding inequality.

Set $K_p = p(2p-1)$ and $F_p = 2p^2$. These are the Kendall's Tau and footrule costs of reversing a block of $2p$ consecutive items. Lemma G.6 relates the ranking distances to the minimum violation-count. Its lower bounds also hold when the output mixes items from different blocks.

Lemma G.6. *For every fixed $p \ge 1$, given a homogeneous strict system $\mathcal{I} = (\langle a_\ell, z\rangle > 0 : \ell \in [m])$ in q variables, the construction above produces $n = 2pm$ pairwise distinct rational items in dimension $d = q+2$ and satisfies*

$$\mathrm{OPT}_{\mathrm{viol}} \le \mathrm{OPT}_{\mathrm{KT}}^{(p)} \le K_p \,\mathrm{OPT}_{\mathrm{viol}},$$
$$2\,\mathrm{OPT}_{\mathrm{viol}} \le \mathrm{OPT}_{\mathrm{FR}}^{(p)} \le F_p \,\mathrm{OPT}_{\mathrm{viol}}.$$

For every fixed p, there exists a polynomial-time algorithm that, given any ranking σ induced without ties by a polynomial score of degree at most p, outputs a rational vector b with $\mathrm{viol}_{\mathcal{I}}(b) \le d_{\mathrm{KT}}(\pi,\sigma)$ and $2\,\mathrm{viol}_{\mathcal{I}}(b) \le d_{\mathrm{FR}}(\pi,\sigma)$. The construction also takes polynomial time for each fixed p, and the coordinates of the constructed items and the returned vector have polynomial-length rational representations.

Proof. *Distinctness and encoding.* Different blocks have different first coordinates. Within a block, the distinct evaluation values distinguish different i, and the two choices of η differ because $a_\ell \ne 0$. All items are therefore distinct, even when coefficient vectors repeat. The numbers t_i, c_i, v_i and the coefficients of g depend only on p. For fixed p, creating the items takes polynomial time in S and gives polynomial encoding length.

Constructing a ranking from inequality weights. Choose weights z that minimize the violation-count. The perturbation and rational feasibility argument in Appendix E.1 let us choose rational z with $\langle a_\ell, z\rangle \ne 0$ for every ℓ, without increasing its violation-count. Put $A = \max_{\ell,i} |v_i\langle a_\ell, z\rangle|$, choose $\lambda = 2A+1$, and use the degree-at-most-p score

$$f(\rho, x, t) = \lambda\rho + g(t)\langle z, x\rangle.$$

At an item in block ℓ, this score is $\lambda(m-\ell) + \eta v_i \langle a_\ell, z\rangle$. For any $\ell < \ell'$, every item in block ℓ has score at least $\lambda - 2A = 1$ greater than every item in block ℓ', so the blocks remain consecutive and follow their target order. If the inequality holds, the order within its block agrees with the target. If it fails, its nonzero inner product is negative, and the order within the block is the reverse of its target order. All scores are distinct. Each failing inequality contributes $\binom{2p}{2} = K_p$ inversions and $\sum_{j=1}^{2p} |2p+1-2j| = F_p$ units of footrule distance. Different blocks create no other errors. This proves the two upper bounds because z minimizes the violation-count.

Obtaining inequality weights from any realizable ranking. Expand the constructed items into their monomial vectors and solve the adjacent-score constraints for the returned order, requiring each score difference to be at least 1. A polynomial score that induces the ranking without ties makes this rational linear program feasible after positive scaling. Its rational solution has polynomial encoding length for fixed p, because the proof of Theorem G.1 bounds the encoding length of the expanded coordinates. Interpret its weights as a polynomial f and obtain b from Lemma G.5. For each inequality occurrence, define $H_\ell = \sum_i c_i[f(m-\ell, a_\ell, t_i) - f(m-\ell, -a_\ell, t_i)]$. Lemma G.5 gives $H_\ell = 2\langle b, a_\ell\rangle$.

Suppose the returned ranking keeps the target order within block ℓ. Every item with positive coefficient ηc_i then has greater score than every item with negative coefficient. Let m_+ be the smallest score in the positive-coefficient group, and let m_- be the largest score in the negative-coefficient group. We have $m_+ > m_-$. The sum of the positive coefficients is $W = \sum_i |c_i| > 0$, and the negative coefficients sum to $-W$. Consequently, $H_\ell \ge W(m_+ - m_-) > 0$. If $\langle b, a_\ell\rangle \le 0$, the output must therefore reverse at least one pair within that block. These pairs belong to distinct blocks, so the violation-count is at most the Kendall's Tau distance.

For footrule, choose one reversed pair from each such block. Suppose its two items occupy target positions $j < k$, while those same items occupy output positions $j' > k'$, respectively. Their contribution is at least $|j'-j| + |k'-k| \ge (j'-k') + (k-j) \ge 2$. Different blocks have disjoint items, so summing these bounds gives twice the violation-count as a lower bound on footrule distance. This argument does not require output blocks to remain consecutive. No resulting weight vector can have fewer violations than $\mathrm{OPT}_{\mathrm{viol}}$, which proves both optimum lower bounds. Solving the rational linear program and extracting the coefficient vector also proves that this conversion takes polynomial time. □

For Kendall's Tau, an α-approximation on this instance gives weights with at most $K_p\alpha\,\mathrm{OPT}_{\mathrm{viol}}$ violations. For footrule, it gives at most $p^2\alpha\,\mathrm{OPT}_{\mathrm{viol}}$ violations, because Lemma G.6 bounds their number by half the returned footrule distance. Both factors are at most $C_p\alpha$, where $C_p = K_p$. Since the degree is fixed, C_p does not grow with the number of inequalities or the feature dimension. The lower bounds below use this fact to absorb the additional factor.

G.5.3 *Approximation hardness and dimension dependence.*

Theorem G.7 now applies the inequality lower bounds used for linear scores. The additional factor C_p changes only their exponent constants. The result allows an algorithm to use every coefficient of degree at most p, rather than just the special polynomials constructed in the preceding proof.

Theorem G.7. *For every fixed integer $p \ge 1$, consider instances consisting of $X = \{x_1, \dots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π of X, where d is part of the input. There are constants $c_p, c'_p > 0$ such that the polynomial minimum-disagreement problems under Kendall's Tau and Spearman footrule distances admit no deterministic polynomial-time $n^{c_p/\log\log n}$-approximation unless* P = NP. *Under PGC, a deterministic polynomial-time $n^{c'_p}$-approximation for either objective would also imply* P = NP.

Proof. Apply Lemma G.6 to the inequality systems in Theorem E.1, whose lower bounds follow from [44]. Let $c_0, c_1 > 0$ be their exponent constants. A ranking approximation of factor α would give an approximation of factor at most $C_p\alpha$ for the violation-count. Both conversions take polynomial time for fixed p, and $n = 2pm \le 2pS$.

For large S, we have $2pS \le S^2$. The ratio $\log t/\log\log t$ eventually increases with t, so

$$\log\left(C_p n^{c_p/\log\log n}\right) \le \log C_p + \frac{2c_p \log S}{\log\log S}.$$

Choose $0 < c_p < c_0/4$. The right side is strictly less than $c_0 \log S/\log\log S$ for sufficiently large S. Thus the resulting approximation factor is smaller than the lower bound $S^{c_0/\log\log S}$, a contradiction. For the PGC bound, choose $0 < c'_p < c_1/2$. Then $C_p(2pS)^{c'_p} < S^{c_1}$ for sufficiently large S, giving the other contradiction. As in the main hardness statements, the factor expressions are asymptotic; changing their values for the finitely many smaller item counts does not affect either contradiction. □

Removing redundant original coordinates must preserve the entire polynomial scoring family, not just its linear part. Substituting linear expressions plus constants preserves the maximum degree and gives the required correspondence. Theorem G.8 expresses hardness both in the supplied feature dimension and in the effective dimension of the expanded features, which controls the algorithms for polynomial scores.

Theorem G.8. *For every fixed integer $p \ge 1$, consider instances consisting of $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π of X. Let r be the dimension of the span of the vectors $x_i - x_j$, and let r_p be the dimension of the span of the vectors $\Phi_p(x_i) - \Phi_p(x_j)$, in both cases over all $1 \le i < j \le n$. There are constants $a_p, b_p > 0$ such that the polynomial minimum-disagreement problems under Kendall's Tau and Spearman footrule distances admit no deterministic polynomial-time $d^{a_p/\log\log d}$-approximation unless* P = NP. *Under PGC, a deterministic polynomial-time d^{b_p}-approximation for either objective would also imply* P = NP. *The same conclusions hold with r_p in place of d, after decreasing the positive constants if necessary. All these bounds hold even when the original ambient and effective dimensions coincide, $d = r$. We interpret the first factors asymptotically, with any fixed bound in the finitely many smaller dimensions.*

Proof. Start with the block instance in $\mathbb{Q}^{q+2}$. Let $u_j = u_1 + Py_j$ express its items in a rational basis, where the matrix P has full column rank r and the differences of the y_j span $\mathbb{R}^r$. Gaussian elimination computes these coordinates in polynomial time, as in the preliminaries. The rational matrix $T = (P^\top P)^{-1}P^\top$ satisfies $TP = I_r$.

Every original polynomial f gives a polynomial $h(y) = f(u_1 + Py)$ of degree at most p on the compressed coordinates. Conversely, every such h extends to $f(u) = h(T(u - u_1))$, also of degree at most p, and $f(u_j) = h(y_j)$ at every input item. These substitutions show that both polynomial families induce exactly the same rankings without ties and have the same optimum values. They also preserve distinctness: $u_i = u_j$ if and only if $y_i = y_j$, because P has full column rank. The compressed coordinates have polynomial encoding length. Given any returned compressed ranking, solve its adjacent-score constraints in the compressed monomial features to obtain a rational polynomial h. Composing h with $T(u - u_1)$ gives a rational polynomial that induces the same ranking on the original block items. We then obtain weights for the original inequalities using Lemma G.6. These operations have polynomial encoding length and running time for fixed p.

Rename the compressed ambient dimension as d; then $d = r$ and $d \le q + 2 \le S + 2$. Its expanded effective dimension satisfies $r_p \le \binom{d+p}{p} - 1 = O_p(d^p)$. For either dimension measure $\xi \in \{d, r_p\}$, we therefore have $\xi \le S^{2p}$ for sufficiently large S. For large ξ, eventual monotonicity gives

$$\log\left(C_p \xi^{a_p/\log\log \xi}\right) \le \log C_p + \frac{2pa_p \log S}{\log\log S}.$$

Choose $0 < a_p < c_0/(4p)$ to make this strictly smaller than $c_0 \log S/\log\log S$ for sufficiently large S. For PGC, choosing $0 < b_p < c_1/(4p)$ gives $C_p\xi^{b_p} \le C_p S^{2pb_p} < S^{c_1}$. Either such dimension-dependent ranking algorithm would thus give an approximation for the violation-count ruled out by Theorem E.1. The factor expressions are again asymptotic, so the finitely many dimensions outside the monotonic range do not affect the conclusion. This proves the claims for both dimension measures on instances with $d = r$. □

The approximation lower bounds also exclude exact optimization for arbitrary dimension. Complementing Kendall's Tau disagreement gives the maximum-agreement consequence, and integrality rules out an FPTAS for each objective. These consequences do not require PGC.

Corollary G.9. *For every fixed integer $p \ge 1$, consider instances consisting of $X = \{x_1, \ldots, x_n\} \subseteq \mathbb{Q}^d$ and a target ranking π of X, where d is part of the input. Unless* P = NP, *no deterministic polynomial-time exact algorithm or deterministic FPTAS exists for minimizing Kendall's Tau distance, minimizing Spearman footrule distance, or maximizing Kendall agreement.*

Proof. An exact algorithm for either minimum distance would contradict Theorem G.7. For every strict ranking, Kendall agreement equals $\binom{n}{2}$ minus Kendall's Tau distance. Optimizing agreement over the same polynomial ranking family would therefore optimize Kendall's Tau distance exactly, giving the same contradiction.

Now suppose one of these objectives admits an FPTAS. Run it with $\varepsilon = 1/(2n^2)$. Every feasible objective is an integer smaller than n^2. For either minimum distance, the returned value exceeds optimum by less than $1/2$; for agreement, the optimum minus the returned value is less than $1/2$. Its integer value must consequently equal the optimum. The FPTAS runs in time polynomial in the binary input size and $1/\varepsilon = 2n^2$, producing the excluded exact algorithm. □

Scope of the logarithmic-dimension claim. For $p \ge 2$, Lemma G.6 controls approximation gaps only up to a degree-dependent constant. Its bounds do not preserve the one-unit clique gap in Theorem E.2 in Appendix E.3. The reduction therefore does not establish exact or FPTAS hardness for higher-degree scores in logarithmic dimension. For $p = 1$, the linear theorem applies directly. For higher

degrees, the hardness bounds concern arbitrary dimension, whereas the agreement PTAS applies when $r_p = O(\log n)$.

H Missing Details from Section 6: Datasets and Additional Results

H.1 Dataset description

- **NBA**: Following prior work on related problems [13], we use the NBA dataset [58], which contains 22,840 tuples describing player statistics across the 1979/80–2022/23 seasons. Each tuple corresponds to a player-season combination, uniquely identified by the PLR attribute, comprising the player's name, age, and team. The default ranking attributes are the player's season-average statistics: PTS, REB, AST, STL, BLK, FG%, 3P%, and FT%. In line with the previous work, to eliminate duplicate rankings, we retain only one player-season record when multiple records have identical values across these attributes. We consider two sources of ground-truth rankings: the MVP ranking and a ranking based on the product of Minutes Played (MP) and Player Efficiency Rating (PER). Computing PER involves a complex formula and requires additional attributes beyond those included in our default set of ranking attributes.
- **Retiring Adult** [20]: The Retiring Adult dataset consists of 1664500 records pertaining to the Income of individuals surveyed in the United States. We use a cleaned and numerically encoded version of the full Adult income dataset. Each record represents an individual and is described by ten attributes derived from demographic, educational, employment, and financial information. Ordinal (and categorical) attributes like Education are encoded numerically to facilitate algorithmic processing. The records are ordered according to the income attribute in non-increasing order, and the Income field is removed from the dataset. The ordering provides us with a ranking in need of an explanation.
- **IIT JEE 2009**: JEE is a yearly entrance exam, the scores of which are used as a criterion for admission to the top engineering colleges in India. The dataset consists of Physics, Mathematics, and Chemistry scores for 384,977 candidates from the year 2009. The dataset also captures demographic information along with the scores themselves. As the ranking is not available for this dataset, we follow two models for producing a viable ranking — the Additive bonus and Multiplicative bonus models from Yan et al. [66]. The dataset has 2 minority groups, SC and ST. We use unit weights for Physics, Mathematics, and Chemistry, but account for "*historical biases*" through an additive/multiplicative bonus. For the additive and multiplicative bonuses, we set the bonus such that the difference between the mean of each minority group and the mean of the majority group is minimized, along with ℓ_2-regularization to prevent arbitrarily large bonus scores. Note that determining the additive/multiplicative bonus is a convex optimization problem that can be efficiently solved. Thus, we generate two datasets with rankings based on the additive bonus model and the multiplicative bonus model. The final dataset only consists of the Physics, Chemistry and Mathematics marks of the candidates.
- **Human Cognitive Performance**: We use the Human Cognitive Performance Analysis dataset [53], obtained from Kaggle. Each tuple represents an individual and records demographic characteristics, lifestyle factors, and measures related to cognitive performance. The dataset contains a reported 80,000 records and 11 attributes: age, gender, sleep duration, stress level, diet type, screen time, exercise frequency, caffeine intake, reaction time, cognitive score, and memory test score. We retain 8 numerical attributes (age, sleep duration, stress level, screen time, exercise frequency, caffeine intake, reaction time, and memory test score) and sort the dataset based on cognitive score in non-increasing order.
- **Synthetic dataset**: The synthetic datasets consist of a collection of 7 datasets. The data records within each dataset follow a uniform distribution within a sphere of radius 100. As the scores in this dataset are not available, we pick a linear scoring function to generate a ranked order and then perturb it using the Mallows model [21, 39], which is defined as follows:

$$P(\sigma \mid \sigma_0, \phi) = \frac{\phi^{d_{\mathrm{KT}}(\sigma,\sigma_0)}}{Z_n(\phi)}$$

where $Z_n(\phi) = \prod_{j=1}^{n} \left(1 + \phi + \ldots + \phi^{j-1}\right)$ and ϕ is called the dispersion parameter. In our experiments, we choose ϕ to correspond to an expected number of inversions. We generate 2 kinds of datasets: (i) varying dimensions (3, 7, 11, 15) when the number of tuples is fixed at 1,000,000 (with ϕ chosen such that the selected permutation has 5% inversions), and (ii) varying dataset size (100,000, 200,000, 400,000, 1,000,000) when the number of dimensions is fixed at 3 (with ϕ chosen such that the selected permutation has 5% inversions in expectation). Thus, overall we generate 7 datasets as combinations of parameters.

H.2 Additional experimental results, tables and figures

H.2.1 Effects of unknown scoring models. In the HCP dataset (Table 10), `MaxFS` and `Sampling` (weighted and unweighted) perform nearly as well as each other, closely followed by the `RankSVM` and `RankNet` methods. `Sampling-CLR` and `LMRC` follow these methods but with a noticeable increase in mean-KT scores. The standard deviations for the different methods show that weighted sampling is the most consistent after `LMRC` (deterministic), followed by the sampling algorithm. `MaxFS` follows closely next. `Approximate-Agreement` does not perform as well due to the smaller values of ϵ that can be targeted within the 30-minute time limit.

Table 9 shows the results for the NBA dataset. Due to the smaller size of the NBA dataset, many approaches are able to complete within the 30-minute timeout. `LS` is able to complete within the timeout and performs the best among the different methods. It is closely followed by `RankSVM`, `RankNet`, and `Weighted Sampling`. `Sampling-CLR` and `MaxFS` compete closely except for the large deviations shown by the `MaxFS` approach. Deterministic `LMRC` shows no deviations but does not perform as well as other methods.

H.3 Explanation using linear models — including `OPT`

We start our experiments with the class of linear functions and run all algorithms on the synthetic dataset with 3 dimensions. The dataset is restricted to 100 rows which helps us efficiently compute OPT using an ILP formulation. The results are presented in

Table 9: Per-method Kendall's Tau statistics on the NBA dataset. Lower values are better. Values are reported over 5 different runs with different seeds.

Method	Mean	Median	Std. Dev.
ls	13206428.4	13205293	1827.9
RankSVM	13214827.4	13215616	2702.1
RankNet	13221703.2	13223065	5292.1
Weighted Sampling	13226311.6	13227459	7520.6
Sampling-CLR	13241848	13243073	2996.1
MaxFS	13245519.4	13237678	23148.7
Sampling	13256424.8	13254931	9961.0
LMRC	14025325	14025325	0
Approx-Agree	18647509.2	16016564	4619031.3
1/2-approx	76758870.4	78414135	31138648.0077

Table 10: Per-method Kendall's Tau statistics on the Human Cognitive Performance dataset. Lower values are better. Values are reported over 5 different runs with different seeds.

Method	Mean	Median	Std. Dev.
MaxFS	240441393	240324386	307709.8
Sampling	240460696.4	240424216	263291.4
Weighted Sampling	240579932.8	240588720	116457.7
RankSVM	242121991.4	242061497	832360.5
RankNet	242443678.2	242301239	1193067.4
Sampling-CLR	249681766.8	248843834	2063174.2
LMRC	250517167	250517167	0
Approx-Agree	314840422	306580969	27043109.5
1/2-approx	609923813.8	613372739	17203349.4

Table 11: Comparison to OPT for small uniform dataset

Method	Kendall-Tau
OPT Local Search, Sampling, MaxFS	286
Weighted sampling	290
Sampling-CLR, LMRC, Approx-Agree	291
RankNet, RankSVM	292
CLR	293
1/2-approx	2143

Table 11. With $n = 100$ and optimum disagreement (KT score) 286, the optimum agreement is $4950 - 286 = 4664$. CLR's disagreement ratio is $293/286 = 1.0245$; Approx-Agree's agreement ratio is $4659/4664 = 0.9989$, which exceeds the $1 - \epsilon = 0.75$ target. The half-approximation's agreement ratio is $(4950 - 2143)/4664 = 0.6018$, which is greater than $1/2$. Thus, the results corroborate our approximation factor of $d + 1$ for Closest-Linear Ranking and better than $1/2$ for 1/2-approx. Approx-Agree was experimented with $\epsilon = 0.25$ and the number of Agreements is 4659 which gives a better ratio than $1 - \epsilon = 0.75$.

I Missing Details from Section 6: Polynomial Experiments

Experimental setup. We extend the comparison in Section 6 to polynomial scoring functions using the representation in Appendix G. For a maximum total degree p, each item x_i is represented by the vector $\Phi_p(x_i)$ of all nonconstant monomials of degrees $1, \ldots, p$. We seek a score $f_a(x_i) = a^\top \Phi_p(x_i)$, with unrestricted real coefficients. The expanded dimension is $D_p = \binom{d+p}{p} - 1$, and $p = 1$ recovers the linear scoring family. The implementation defaults to quadratic scores, but supports any fixed degree subject to memory limits. All methods in a comparison use the same degree, input scaling, and monomial representation. Pairwise losses use $v_{ij} = \Phi_p(x_i) - \Phi_p(x_j)$, with $i \succ_\pi j$, and margin $m_{ij} = a^\top v_{ij}$.

Baselines. We compare against seven baselines restricted to this same polynomial scoring family. *Polynomial Linearized MRC* applies the ridge-regularized, linearized rank-correlation estimator of Shen et al. [55] to the expanded features. *Polynomial RankSVM* minimizes the regularized pairwise hinge loss $\max\{0, 1 - m_{ij}\}$ [32]; polynomial-kernel ranking models are also considered in prior work [52]. *Polynomial RankNet* instead minimizes the regularized pairwise logistic loss $\log(1 + \exp(-m_{ij}))$ [8], with the score restricted to f_a, rather than a neural network. *Polynomial RankRLS* uses the preference-graph least-squares formulation of Pahikkala et al. [47]: it minimizes the mean squared preference error $(1-m_{ij})^2/2$, together with an ℓ_2 penalty. Each preference has unit target margin, so the method requires only the target order, not numerical target-score differences. Its matrix-free conjugate-gradient implementation avoids constructing the full pair matrix. *Smoothed MRC* maximizes the mean of $s(m_{ij}/h)$, where $s(t) = 1/(1 + \exp(-t))$, over a unit coefficient direction. It uses multiple starts, projected ascent, and decreasing bandwidth h. This is a logistic-smoothing adaptation motivated by smoothed rank-correlation estimation [67], not a reproduction of its self-induced smoothing procedure. Unlike RankNet, it maximizes sigmoids rather than log-sigmoids. *Polynomial Kendall local search* applies the full-data coordinate and Gaussian search from Section 6 to polynomial coefficients [42]. Finally, *Polynomial MaxFS* applies elastic comparison-slack minimization, greedy deletion, and add-back to the lifted constraints [14]. The pairwise learners use all eligible comparisons when practical, or a bounded uniform sample of original item pairs. RankRLS also supports the full preference graph implicitly, while full-data RankSVM computes hinge subgradients without enumerating pairs. Every baseline returns a polynomial score; agreement and disagreement are recomputed on the full comparison population.

Details of our algorithms. The polynomial feature representation lets us reuse the comparison algorithms from the linear setting. We distinguish the full-instance minimum-disagreement approximation from the two sampled local-search variants used for scalable maximum agreement.

Closest Polynomial Ranking. We run Closest Linear Ranking on all eligible lifted comparisons. The algorithm repeatedly removes an inclusion-minimal infeasible subsystem until the retained comparisons admit a common coefficient vector, then refines its score order while preserving those comparisons. Theorem G.2 gives an $(r_p + 1)$-approximation, where r_p is the dimension spanned by the lifted feature differences. Thus the factor depends on the expanded comparison space, with $r_p \le D_p$, not simply on the original number of attributes d. This method uses the full comparison population and is intended for instances on which the LP and infeasible-subsystem computations are practical.

Sampling. We sample eligible original item pairs uniformly with replacement and run the circle-sweep local search from Section 6 on their lifted differences. Repeated comparisons retain their sampling multiplicities. A thin singular value decomposition restricts the search to the span of the sampled differences, whose dimension is

at most the number of distinct sampled comparisons. Starting directions include the polynomial Linearized MRC pilot, the weighted sample centroid, their reverses, and random directions. Each sweep searches the arcs between sampled comparison sign changes on a great circle and accepts only a numerically verified non-worsening move. Finalists and their negatives are evaluated on the full dataset.

`Weighted Sampling`. To reduce repeated sampling of comparisons between large duplicate classes, we group identical feature rows and use a mixture of the original item-level distribution, a class-balanced distribution, and a distribution favoring small-margin comparisons under the polynomial Linearized MRC pilot. The default mixture weights are 0.5, 0.25, and 0.25, respectively. For a directed class comparison e, let W_e count its eligible original item pairs and let $P = \sum_e W_e$. Its importance weight is proportional to $(W_e/P)/q(e)$, where $q(e)$ is its proposal probability. Thus the weighted search targets original-item agreement rather than giving each class comparison equal weight. Small instances use an explicit class-pair table; larger instances compute multiplicities only for proposed class pairs, avoiding quadratic storage in the number of distinct rows. We then use the same sampled circle-sweep search and full-data finalist selection as `Sampling`.

Both sampled variants are heuristics, not implementations of the exact empirical optimizer required by the polynomial agreement scheme. A fixed sample budget controls the search problem; ε and δ provide reference sample-size diagnostics rather than a PTAS guarantee. The default search uses eight restarts and at most 250 iterations per restart. Comparing each finalist with its reverse nevertheless retains the $1/2$ maximum-agreement guarantee: the two orientations partition the optimizable target comparisons, while target-strict duplicate pairs are unavoidable disagreements in either orientation.

Duplicates, ties, and evaluation. We preserve original-item multiplicities throughout training and evaluation. Target-tied pairs are excluded, identical input rows remain tied, and a target-strict pair of identical rows contributes one forced disagreement. Including degree-one monomials ensures that polynomial lifting does not merge distinct input rows merely because their higher powers coincide. Numerical score ties between distinct lifted point classes are refined and the resulting order is checked before evaluation. We report full-data KT disagreement, agreement, and runtime including feature construction. Sampling reduces the comparison-search cost, but the implementation still constructs the $n \times D_p$ feature matrix for the pilot and final evaluation. Products and optimization use double precision; numerical certificates are checked on the computed lifted features, rather than on exact symbolic polynomial evaluations.